

significant challenge.

Formation of scatternets

There has been a comprehensive study to propose an efficient formation of connected scatternet. One important problem that exists with scatternet as well as piconet formation is to keep track of the device that comes in and goes out of the network—since these are low-power handheld devices with limited communication range over Bluetooth. A device connects to another device at random, according to their 48-bit Bluetooth addresses and clocks, which control the hopping behaviour in inquiry or inquiry scan states. Most protocols seek to reduce the scatternet formation time and form fast routing algorithms.

It turns out that a good scatternet formation protocol should:

1. Be fully distributed and rely on local information
2. Generate connected scatternets
3. Be resilient to the disconnection of nodes anywhere in the scatternet
4. Provide multiple routes for robustness and be self-healing
5. Limit the number of bridges
6. Limit the number of roles a device can assume
7. Be aware of device resources

While forming a scatternet, keep in mind that you are dealing with, in general, small, energy- and processing-power-starved devices. The following points should be taken care of while developing scatternet formation protocols:

1. Minimise scatternet construction time
2. Minimise the amount of control data transmitted
3. Minimise the number of hops required for communication between devices, in order to improve response times
4. Minimise power consumption. Master-and-bridge mode of operation requires more power than slave mode
5. Minimise the number of roles assigned to nodes
6. Minimise the number of piconets to provide faster routing and reduced

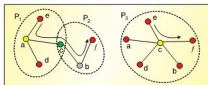


Fig. 2: Showing efficient network after role switching

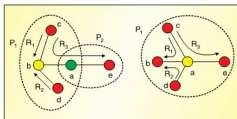


Fig. 3: Unnecessary bridge elimination

packet collisions

7. Maximise the amount of data throughput

Given a scatternet, you can evaluate its performance based on the number of piconets, the number of nodes per piconet, the number of bridge nodes, the number of roles per node, average traffic delay, throughput and maximum traffic delay.

Numerous models have been proposed for constructing Bluetooth scatternets, some of which may require modifications to the existing Bluetooth specifications.

Role switching

Role switching enables two devices to exchange roles very rapidly, rather than reconnecting by executing the time-consuming inquiry and inquiry scan processes. The role switching operation involves fewer slots than the inquiry/inquiry scan and page/page scan operations in switching the roles of the devices.

There are three major types of role switching operations:

1. Combining. A situation where a single node works both as slave in one piconet and master in another.
2. Splitting. A piconet with a large number of devices divided in two piconets with smaller number of devices.
3. Take-over. When the existing master or a bridge is about to move out

of the scatternet, it transfers its responsibilities to another device in the scatternet.

There is a switching delay (called 'guard time') introduced by bridges while they try to transmit packets across the piconets. Eliminating the

unnecessary bridges from the network will improve the performance of the scatternet. There exist various protocols and algorithms to optimise the roles that a device assumes in a scatternet.

We see in the following examples how effective role assignment can reduce the hop count while routing a packet, help reduce the number of bridges in the network and result in an efficient scatternet structure.

In Fig. 2, the original network was formed with 'c' as a bridge node and 'a' and 'b' as the masters on two different piconets p1 and p2. Using role switching, with the help of 'c', improves the structure of the scatternet (now formed as a 'piconet'), shortens the routing path and eliminates the bridge delay in the network.

Fig. 3 shows another example of better role switching operation that results in an improved network structure.

Scatternet applications

By now you might be wondering what possible use these scatternets could be put to. Well, there are papers proposing voice communication between two mobile devices connected over a Bluetooth scatternet, up to a distance of 100 metres in an indoor environment. One of the popular applications developed by Nokia, called 'sensor', works on somewhat similar principles. It lets users communicate to each other, bypassing the service provider's network, over Bluetooth.

Sensor networks are another example where the Bluetooth scatternets can be used. Since sensor networks are generally *ad-hoc*, Bluetooth scatternet can be used for communication of sensors with each other and the base. This will eliminate the requirement of